



August - September 2026 | Women In Data™ Datathon | What's Cooking?

HerPlate Ghana: Diagnosing Food System Breakdowns on Women's Plates

Track: Eat

Presented by: Plateful of Data



Our Team



Manogna R

Food Systems Strategist



Cristiane Pereira

Agrifood Data Analyst



Syeda Nowshin Ibnat

Nutrition Informatics
Engineer



Lucy Sophia

Interactive Tools &
Decision-Support
Specialist

Table of Contents

1

Problem Statement and Objectives

2

Methodology and Analytical Workflow

3

Key Findings

4

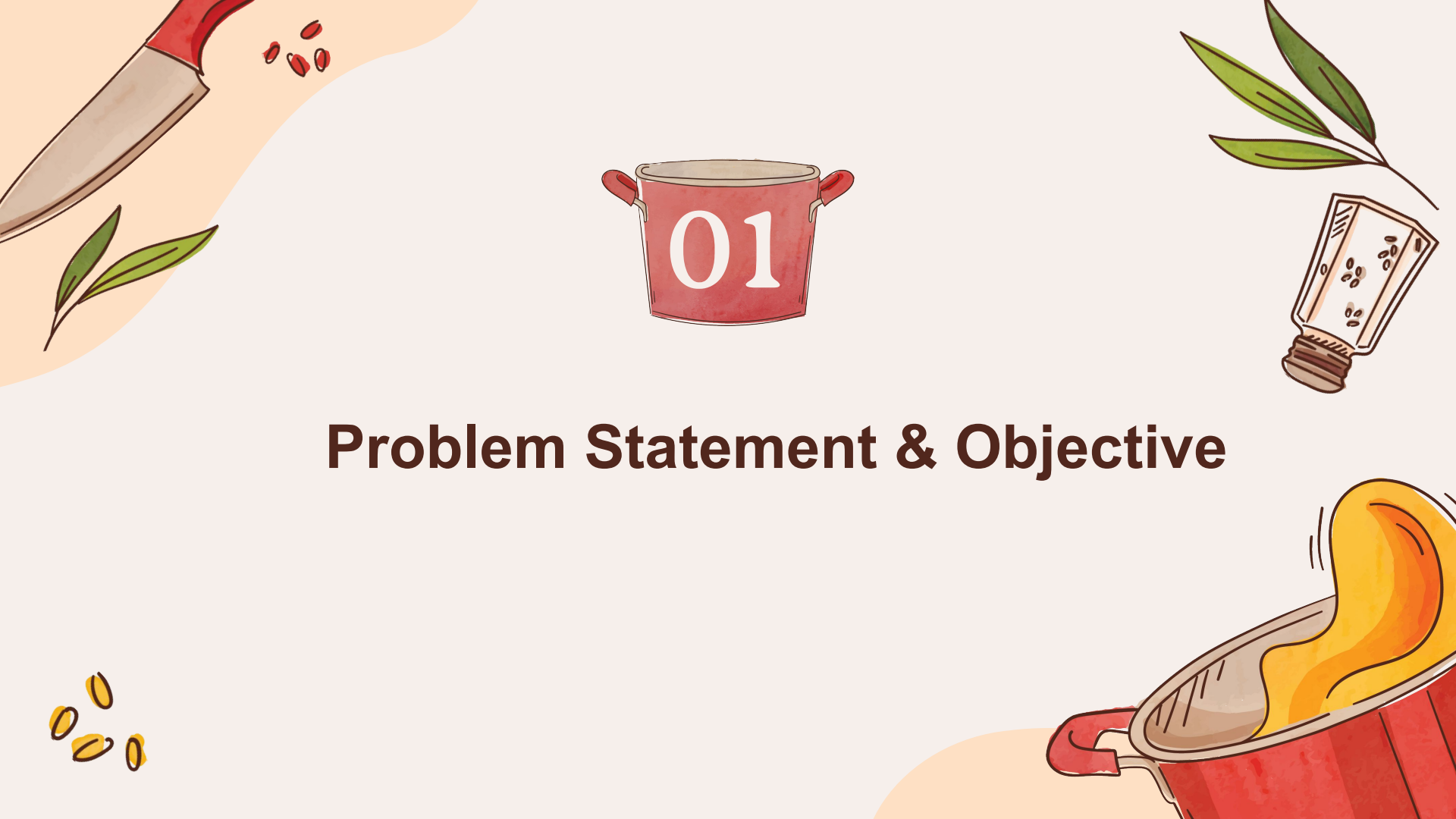
Solution Highlights

5

Actionable Recommendation and Roadmap

6

Conclusion



Problem Statement & Objective

CAN A HEALTHIER PLATE BECOME AN AFFORDABLE PLATE?

Globally Around **2.69 billion people (32.7%)** worldwide are unable to afford a healthy diet.

About **66.6% of Africa's population**—just over **1 billion people**—could not afford a healthy diet in 2025

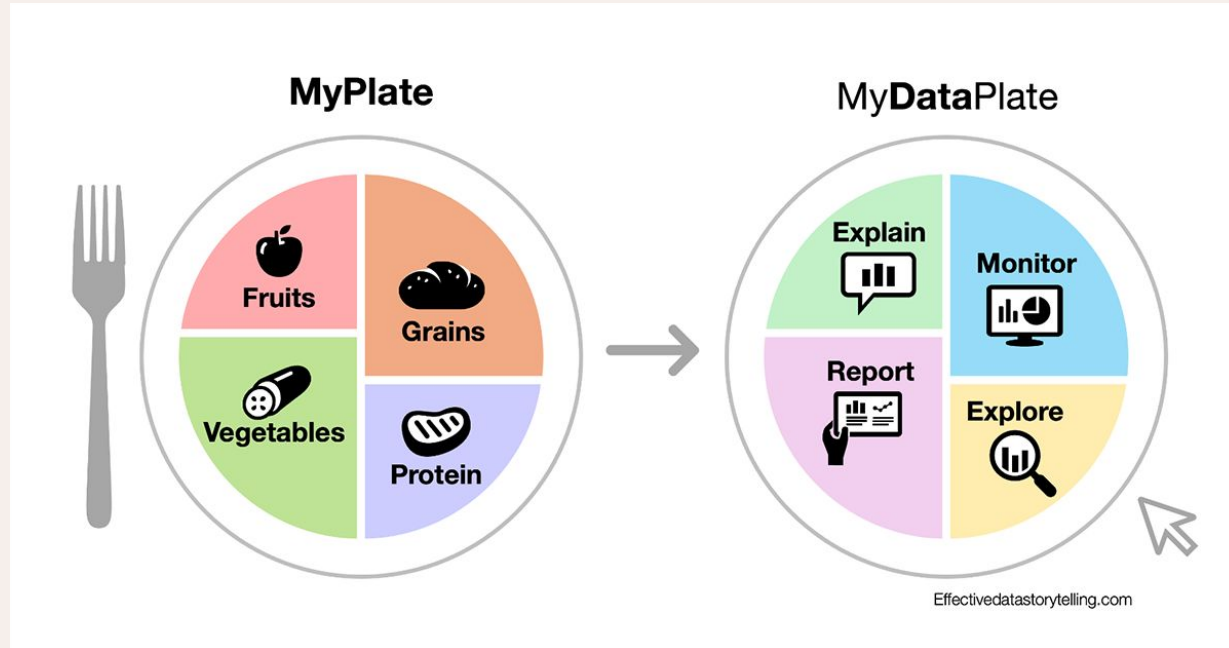
In Ghana, only **49.9%** of women achieve Minimum Dietary Diversity (MDD-W)—meaning over half fail to consume foods from at least 5 of 10 essential food groups daily.



Before the first grain takes root, the farmer ploughs through the muddy fields under a bright sky, while white egrets trail behind, feeding on the insects that have been stirred up. This image reveals how food begins, shaped by those who work the ground. In each motion, there is strength, balance, and deep respect for the life the earth sustains. Photo:©Writchid Snipes

What is on the plate depends on more than nutrition knowledge.

What would it take to make a healthy, diverse plate both available and affordable in Ghana?"



Objectives

Examine food systems from "farm to fork" using real-world UN FAOSTAT data to identify opportunities that strengthen food security, nutrition, and population health.

Connecting Supply to the Plate: Link national agricultural food availability (the "farm" supply) directly to individual dietary diversity and household diet affordability (the "fork").





Your Health • Our Concern

STAKEHOLDERS



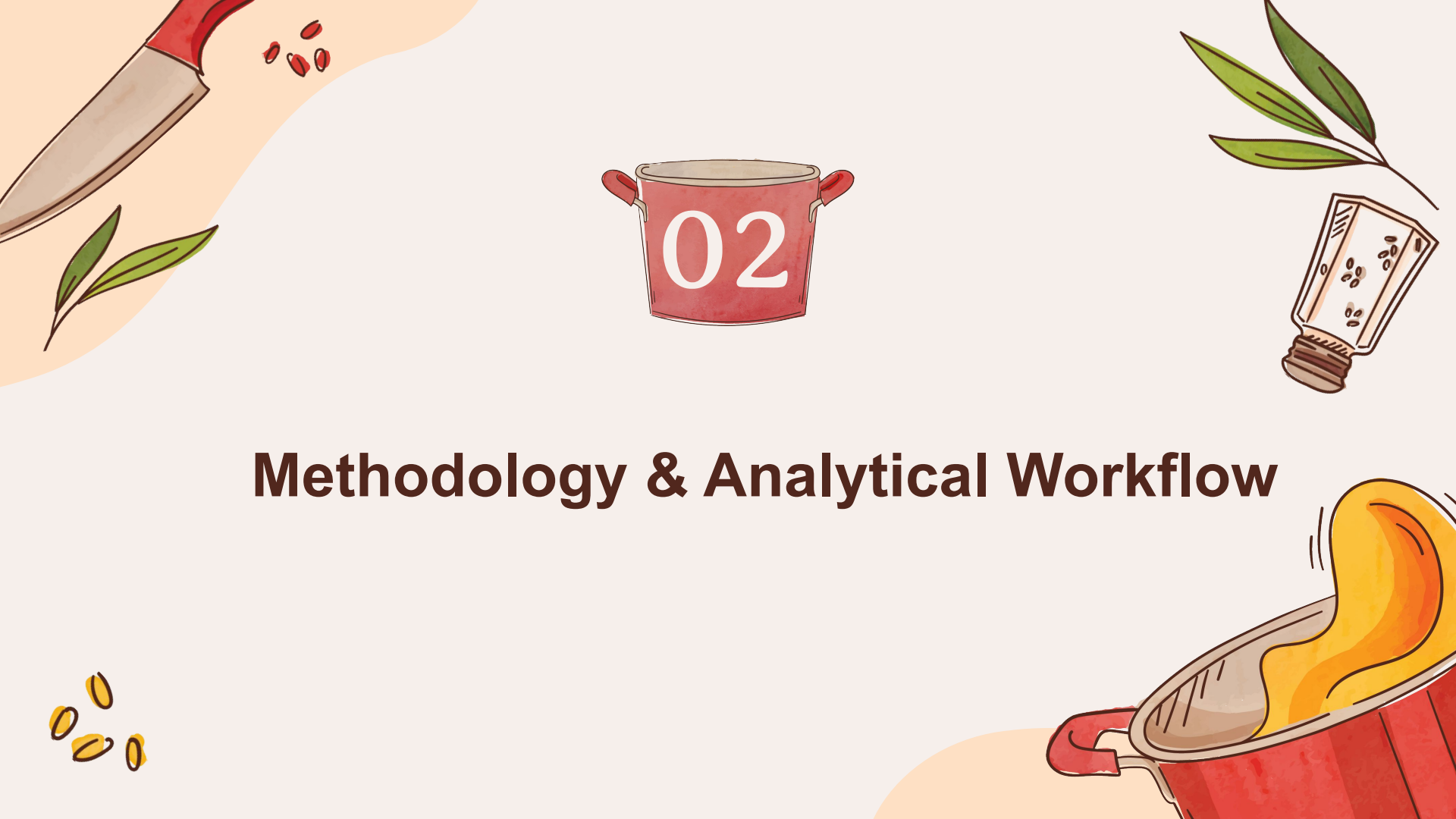
MINISTRY OF
FOOD & AGRICULTURE
REPUBLIC OF GHANA



Republic of Ghana

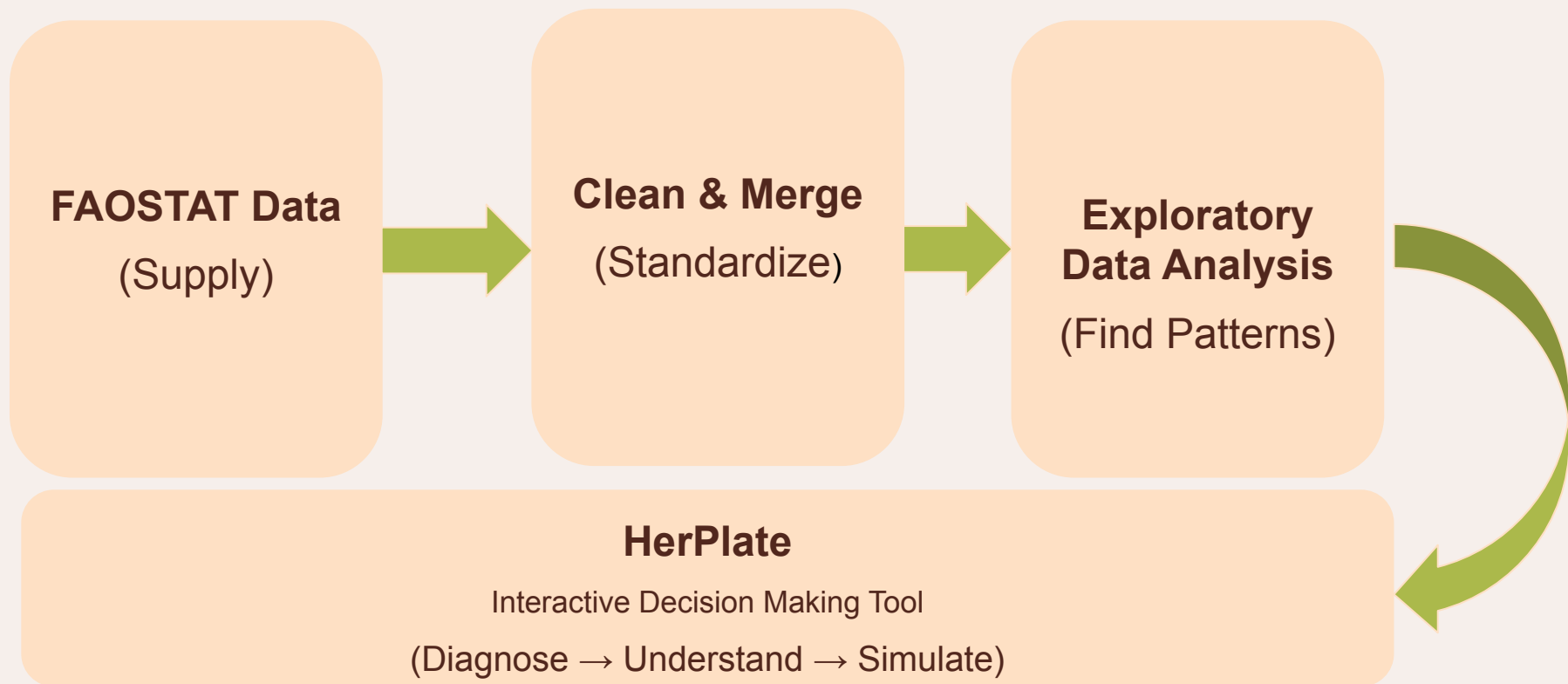


Stakeholder Group	Key Organizations	Internal Stakeholder	External Stakeholder	Priority Stakeholder
Policymakers — Food & Nutrition Policy	Ministry of Food and Agriculture (MoFA), National Development Planning Commission (NDPC)	✓		
Public Health Organizations — Maternal Nutrition Programs	Ghana Health Service (GHS), Ministry of Health (MoH), National Nutrition Technical Committee	✓		✓
Supply-Chain Planners — Crop & Livestock Planning	National Poultry & Egg Producers, Regional Cold-Chain Operators, Grain & Legume Aggregators		✓	



Methodology & Analytical Workflow

From Farm to Fork: Our 4-Step Framework



Tools

Data Acquisition

FAOSTAT CSV
exports



Cleaning, Processing, EDA and Visualization

Python (pandas, numpy,
Matplotlib / Plotly), Jupyter
Notebook, Google Colab, VS
Code



Interactive Web Application

Streamlit, VS Code



Version Control System - Git



Three FAOSTAT Datasets, One Story

Dataset	Period	What it tells us
 MDD-W	Ghana 2022 survey	What women actually consume
 Food Balances	2010-2023	National food supply trends
 CoAHD	2017-2025	Cost and affordability of healthy diets

 Food Balances

 Cost and Affordability of a Healthy Diet

 Food and Diet

Cleaning and Processing

Filtered
Ghana-specific
records across all
three datasets.



Standardized
food-group mappings
(30+ individual items
→ 10 groups)



Validated year
coverage
(2010–2023 for Food
Balances).



Handled missing
values and unit
inconsistencies.

Feature Engineering and EDA

Computed 10 years % change in supply (2012 - 2022) by food group

Cross-referenced supply trends with consumption patterns

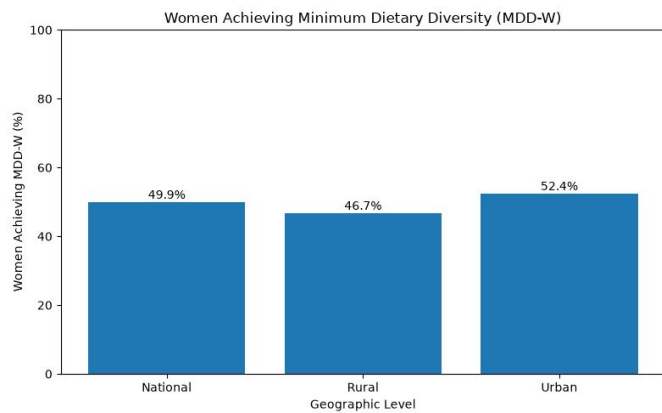
Calculated rural-urban consumption gaps for each MDD-W food group

Analyzed healthy diet basket cost shares and macro inflation trends (CoAHD)

```
# Feature Extraction: 10-Year Availability Percentage Shifts (2012 -> 2022 Anchor)
#  $\Delta S_i = ((S_{2022} - S_{2012}) / S_{2012}) * 100$ 
base_2012 = df_supply_ts[df_supply_ts['Year'] == 2012].iloc[0]
curr_2022 = df_supply_ts[df_supply_ts['Year'] == 2022].iloc[0]

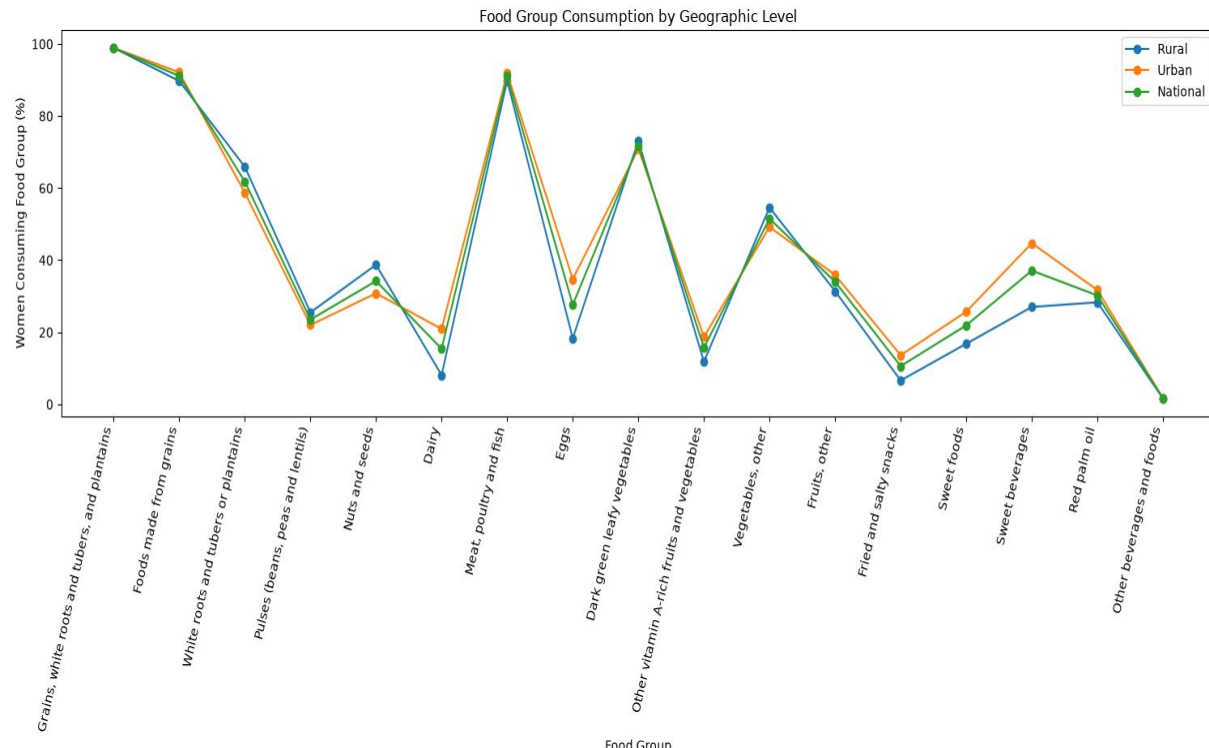
delta_records = []
for col in ['Dairy', 'Eggs', 'Pulses', 'Vegetables']:
    val_2012 = base_2012[col]
    val_2022 = curr_2022[col]
    pct_shift = ((val_2022 - val_2012) / val_2012) * 100
    delta_records.append({
        'HerPlate_Group': col,
        'Supply_2012_kg_cap_yr': np.round(val_2012, 2),
        'Supply_2022_kg_cap_yr': np.round(val_2022, 2),
        'Ten_Year_Delta_Pct': np.round(pct_shift, 1)
    })

df_deltas = pd.DataFrame(delta_records)
```

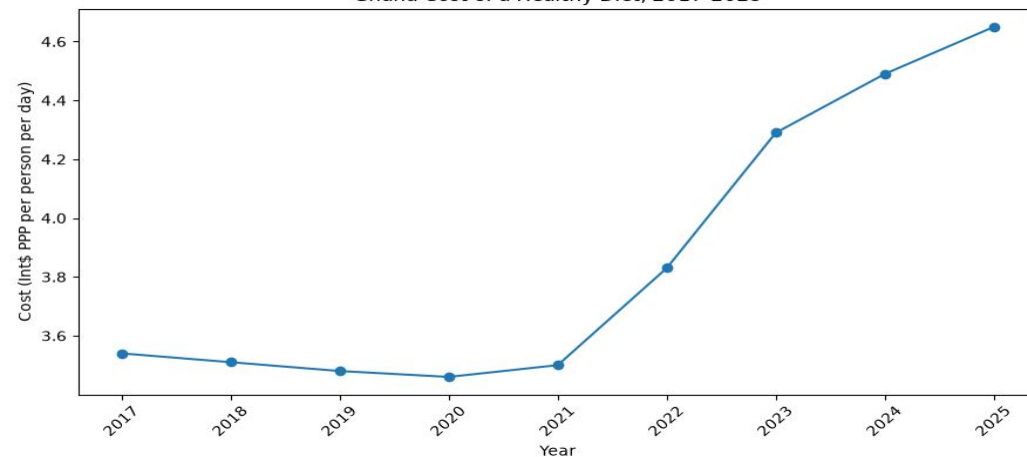


➔ **49.9%** Woman achieve Minimum Dietary Diversity (MDD-W) Nationally.

Food supply shift
2010-2023: pulses ↓
39%, meat ↑ **32%**



Ghana Cost of a Healthy Diet, 2017-2025

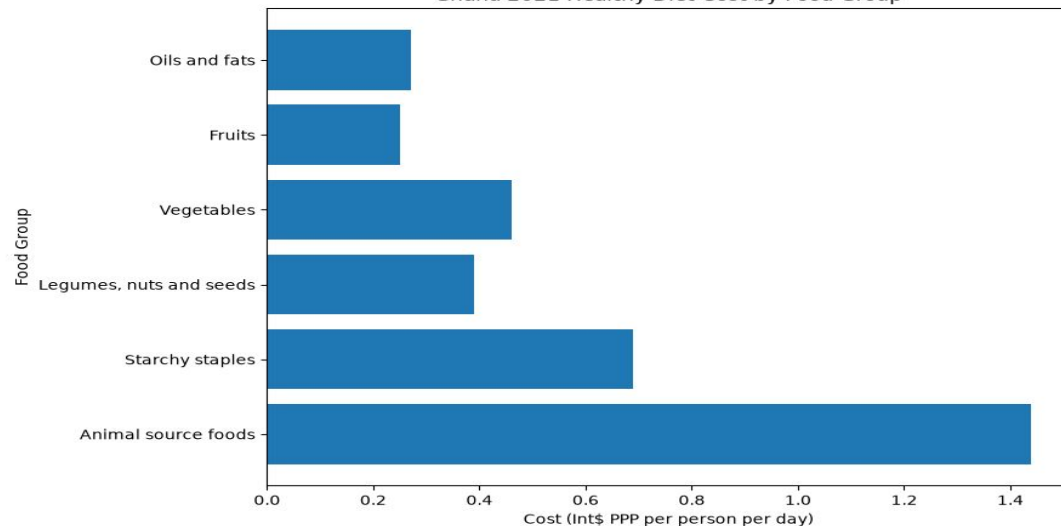


Healthy diet cost rose from \$3.54 to \$4.65 (2017-2025)

Animal-source foods dominate the cost of a healthy diet in Ghana at \$1.44 PPP per person per day, creating the primary financial barrier to dietary adequacy.



Ghana 2021 Healthy Diet Cost by Food Group



Exploratory Data Analysis (EDA)

Obstacles

Hypothesis Pivot

Pivoted from linear programming cost-minimization to bottleneck diagnostics because low intake stems from distinct structural supply and access failures rather than simple food substitution.

Disparate Data Granularities

Reconciled cross-sectional surveys, decadal time series, and commodity price datasets by treating them as complementary signals rather than directly equivalent variables.

Food Balances vs. Consumption

Paired aggregate food balance availability with granular MDD-W survey data to decouple national production volumes from plate-level intake deficits.

Simulation Constraints

Explicitly framed all app simulations as illustrative policy scenarios to prevent macroeconomic projections from being misinterpreted as definitive predictive forecasts.

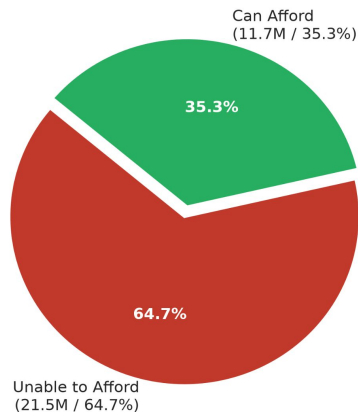


Key Findings — The 4 Bottleneck Archetypes

Key Findings

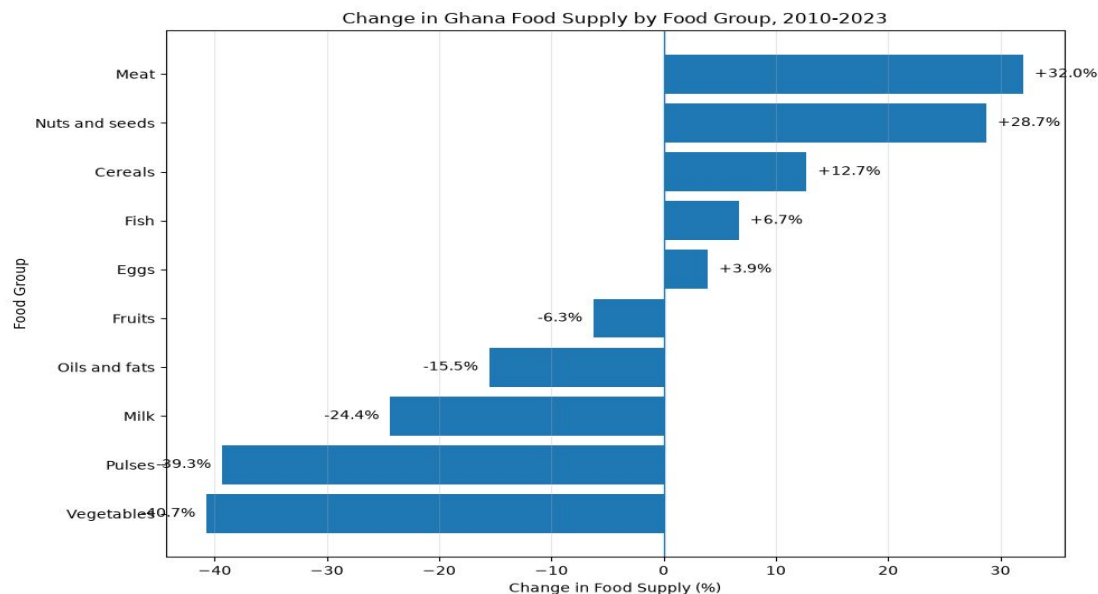
- Low Diversity Baseline: Only 49.9% of Ghanaian women achieve Minimum Dietary Diversity (MDD-W).
- Divergent Supply Shifts: Per-capita meat supply grew +32.0%, while vegetables (-40.7%) and pulses (-39.3%) dropped drastically.
- Urban-Rural Disparity: Egg intake reveals a 16.4 percentage point gap between rural (18.3%) and urban (34.7%) women.
- Access vs. Supply Bottlenecks: Stable egg production (+3.9%) indicates economic access barriers rather than supply shortages.
- High-Impact Scenario: Shifting 50% animal protein to plant sources boosts dietary iron by +55.2% and reaches ~2.2M additional people by lowering diet cost.

Ghana Population Diet Affordability Divide (2022 MDD-W Baseline)



At the 2022 survey baseline, **64.7% of Ghanaians (21.5 million people) were priced out of a healthy diet**, with high animal-source foods creating the primary financial barrier.

National protein availability has increasingly relied on cereals and meat, while **critical plant proteins like pulses steadily stagnated and collapsed per capita**.



Bottleneck Overview

Ghana does not appear to have one single nutrition bottleneck. Different foods show different types of pressure.



Dairy

Convergent Pressure

women's consumption is very low (~15.4%), milk availability declined ~24.4%, and dairy belongs to the animal-source-food category, which is the largest component of the healthy-diet basket (~41.1%).



Pulses

Supply/Diversity Pressure

consumption is low (~23.5%) while national availability declined ~39.4%, but the parent healthy-diet category is not as expensive as animal-source foods.



Eggs

Access Beyond Supply

consumption is low (~27.6%), but national egg availability is roughly stable (+3.9%). This suggests that declining aggregate supply alone cannot explain why eggs are not reaching women's plates



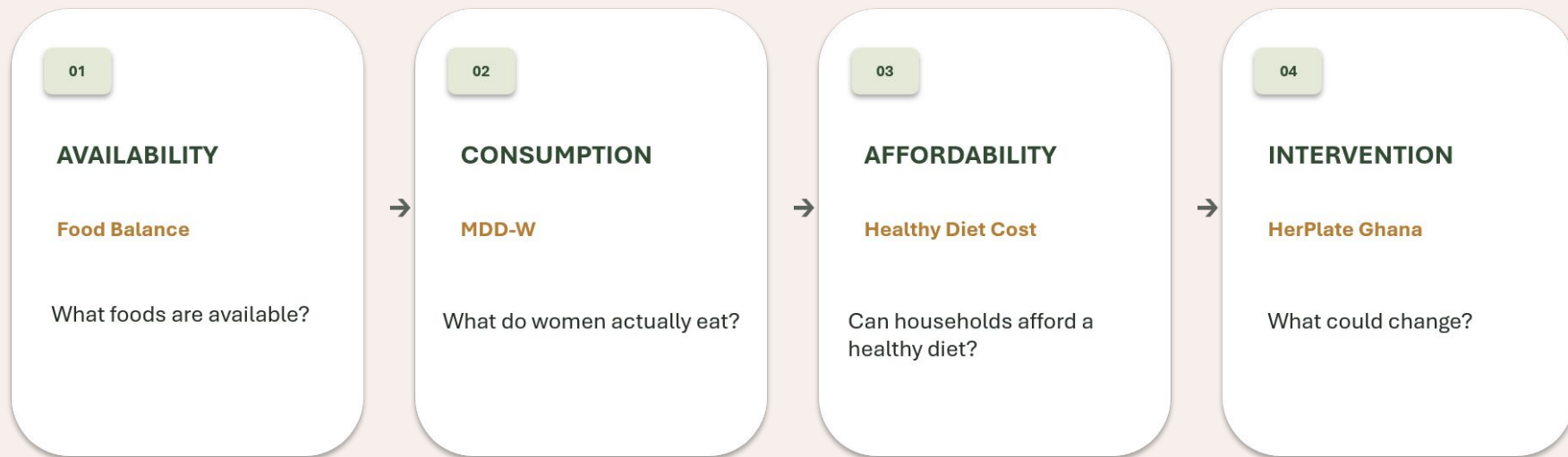
Vegetables

Emerging Supply Concern

consumption is stronger than dairy/eggs/pulses, but national vegetable availability has fallen sharply (~40.7%), making this a potential upstream risk.

"Same dietary gap. Different bottlenecks."

The Problem is Bigger than The Plate



Four signals help us move from describing the problem to exploring where it may break.



Solution Highlights — HerPlate Ghana

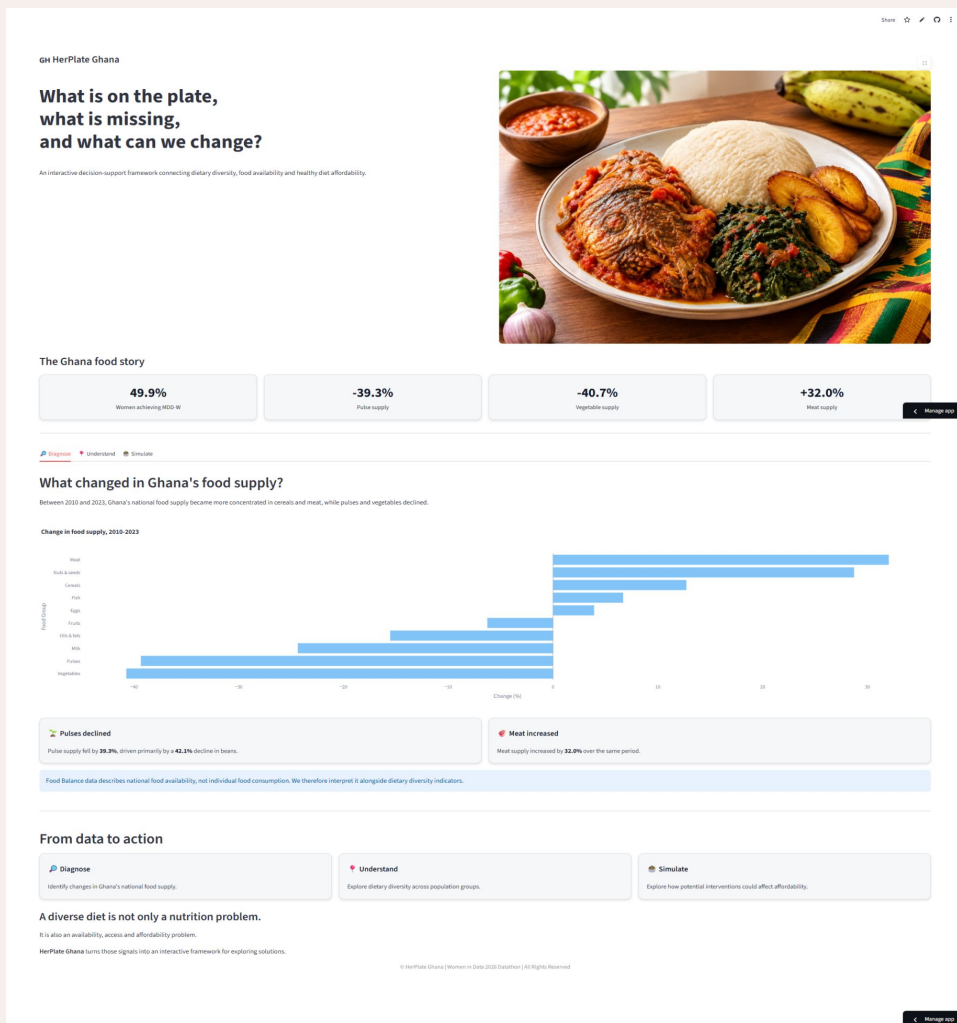


Meet “HerPlate Ghana”

We are not just asking "Which foods do women eat less?"
We are asking whether low consumption coincides with high cost, declining national supply, or a gap that remains even when national supply is stable.

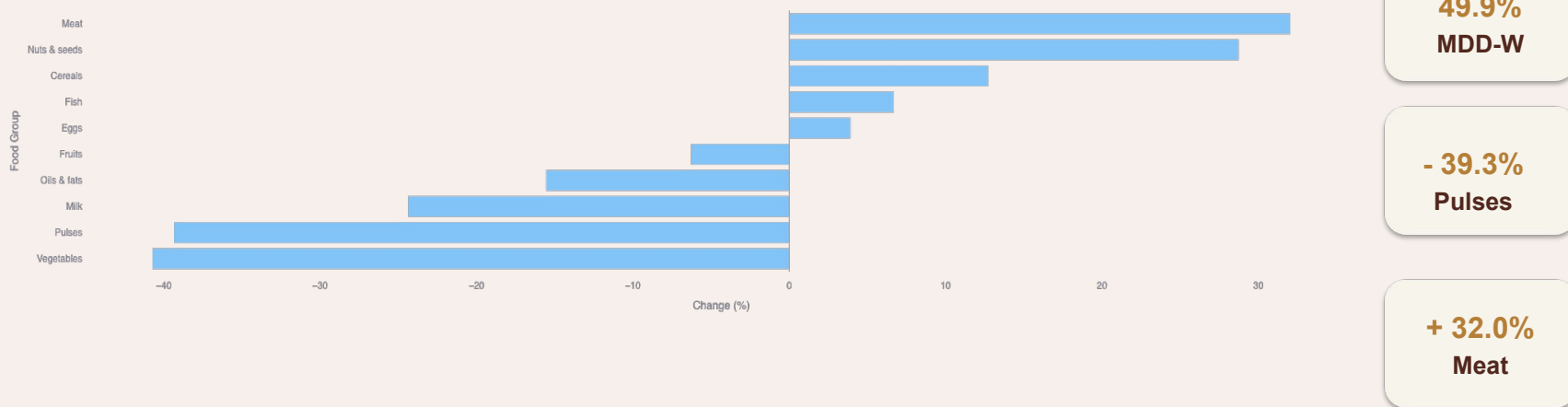
Checkout Our Interactive Tool:
[HerPlate Ghana](#)

*From describing the problem to
exploring possible solutions.*



Diagnose: What Changed in Ghana's Food Supply

Change in food supply, 2010-2023



- Between 2010 and 2023, Ghana's national food supply became more concentrated in cereals and meat, while pulses and vegetables declined.
- Pulse supply fell by 39.3%, driven primarily by a 42.1% decline in beans.
- Meat supply increased by 32.0% over the same period.

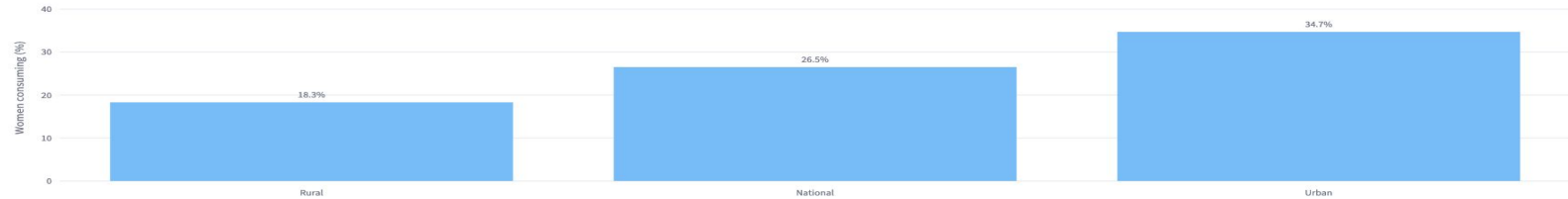
Understand: Who Eats What?

Dietary diversity differs across geographic groups. Explore how consumption varies between rural and urban women.

Select a food group

Eggs

Eggs consumption by geographic level



The rural-urban gap

For **eggs**, the urban consumption rate is **16.4 percentage points** higher than the rural rate.

A closer look at eggs

Rural women

18.3%

Urban women

34.7%

Urban-rural gap

+16.4 pp

MDD-W measures whether women aged 15-49 consumed foods from at least 5 of 10 core food groups.

Egg consumption is almost twice as high among urban women as rural women

Simulate: What Could We Change?

A 50% protein substitution illustrates the potential to increase plant protein while reducing healthy-diet costs from \$4.28 to \$3.75 PPP/day.





Actionable Recommendations

Three Actions for a Healthier Food System

Provide quality seeds and fertilizer to local cowpea and bean growers to reverse the drop in national harvest and bring back cheap plant protein.

Target rural nutrition programs — close the urban–rural gap in eggs, dairy, and fruits.

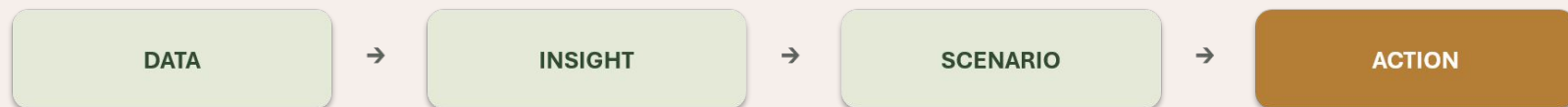
Unify Ministry of Agriculture wholesale price feeds with survey data to enhance coordination among farmers, traders, and public health agencies.



Conclusion



A diverse diet is not only a nutrition problem, it is an availability, access, and affordability problem.



HerPlate Ghana turns those signals into an interactive framework for exploring solutions.

The question is no longer only,
What should be on the plate?

“What would it take to make that plate possible?”



Thank you so much, Women in Data!

This community is continuously helping us
to connect, grow and build together.



Explore our work:

[Women in Data Datathon Project 2026](#)

CREDITS: This presentation template was created by **Slidesgo**, and
includes icons by **Flaticon**, and infographics & images by **Freepik**



Sources

- 1) Food and Agriculture Organization of the United Nations (FAO), "Cost and Affordability of a Healthy Diet (CoAHD) — Ghana," *FAOSTAT*, 2026. [Online]. Available: <https://www.fao.org/faostat/en/#data/CAHD>
- 2) Food and Agriculture Organization of the United Nations (FAO), "Minimum Dietary Diversity for Women (MDD-W) — Ghana," *FAOSTAT*, 2022. [Online]. Available: <https://www.fao.org/faostat/en/#data/MDDW>
- 3) Food and Agriculture Organization of the United Nations (FAO), "Food Balances (2010-) — Ghana," *FAOSTAT*, 2026. [Online]. Available: <https://www.fao.org/faostat/en/#data/FBS>
- 4) FAO, *The State of Food Security and Nutrition in the World 2026*, Rome: FAO, 2026. [Online]. Available: <https://ipes-food.org/un-hunger-report-2026-global-hunger-remains-stubbornly-high-as-another-food-price-crisis-looms/>
- 5) Action Against Hunger, "SOFI Report: Hunger remains at alarming levels as Africa becomes global epicentre," 2026. [Online]. Available: <https://www.actioncontrelafaim.org/en/news/press/sofi-report-hunger-remains-at-alarming-levels-as-africa-becomes-the-global-epicentre/>
- 6) AP News, "UN agency warns cost of a healthy diet has spiked 25% since 2021," *NZ Herald*, 2026. [Online]. Available: <https://www.nzherald.co.nz/world/un-agency-warns-cost-of-a-healthy-diet-has-spiked-25-since-2021/5NPHYZ4NE5A4NPRTBLW66VWVJU/>
- 7) Y. Martin-Prevel *et al.*, "Moving forward on measuring women's diet quality globally: Results from the Women's Dietary Diversity Project," *Matern. Child Nutr.*, vol. 12, no. 1, pp. 1–14, 2016. [Online]. Available: <https://doi.org/10.1111/mcn.12243>